

BRG-Bokeh: Boundary-Refined 3DGS Segmentation for Multi-View Consistent Bokeh

Supplementary Material

In this supplementary material, we provide additional implementation details and comprehensive experimental results to support the main paper. First, we elaborate on the implementation configurations in Sec. VI. Then, we present further quantitative and qualitative evaluations of our boundary-refined segmentation in Sec. VII. Sec. VIII provides a detailed ablation study on bokeh rendering, analyzing the impact of segmentation quality, rendering strategies, and parameter sensitivity. Finally, we report the computational efficiency in Sec. IX and demonstrate the text-driven focus control capabilities in Sec. X.

VI. Implementation Details

Our method is implemented in PyTorch. The segmentation component is configured following the specifications outlined in Sec. III-C. For depth estimation, we employ the DA3-Large variant of Depth Anything v3 [20] to generate high-quality depth maps. Detailed hyperparameter settings for each module are provided in the methodology sections. All experiments were conducted on a single NVIDIA RTX 3090 GPU.

VII. More Segmentation Results

To comprehensively evaluate the visual fidelity of the segmented foreground objects in the absence of ground-truth reference images, we employ CLIP-IQA [41], a no-reference metric that quantifies the alignment between the rendered image and specific textual descriptions. As presented in Table IV, we benchmark performance using three antonym prompt pairs focusing on boundary quality (Clarity, Smoothness) and object integrity (Completeness).

TABLE IV
QUANTITATIVE VISUAL RESULTS ON NVOS DATASET. WE REPORT CLIP-IQA SCORES ($\uparrow$) COMPARING OUR METHOD WITH STATE-OF-THE-ART APPROACHES. OUR METHOD ACHIEVES THE BEST PERFORMANCE IN BOTH BOUNDARY QUALITY AND OBJECT INTEGRITY PROMPTS.

Method	Boundary Quality		Object Integrity
	Clarity (Clear/Unclear)	Smoothness (Smooth/Noisy)	Completeness (Complete/Mutilated)
SAGD [30]	0.5282	0.6510	0.4116
SA3D-GS [31]	0.5597	0.7408	0.4362
COB-GS [23]	0.5801	0.7542	0.4487
Ours	0.5815	0.7585	0.4695

Quantitative results reveal that feature-based methods like SAGD [30] and SAGA [32] suffer from significant degradation in visual quality, exhibiting notably low scores in smoothness. While SA3D-GS [31] improves upon these baselines, it still exhibits obvious ambiguous boundary Gaussians, resulting in suboptimal clarity scores. This quantitative trend is explicitly corroborated by the qualitative comparisons in Fig. 4. Specifically, as observed in the “fern” and “horns_left” scenes, SAGA struggles with high-frequency background clutter due to feature ambiguity, resulting in noticeable floating noise and unclear boundaries. Furthermore, in the “flower” scene, it fails to preserve object integrity, exhibiting significant erosion where petal structures are missing, which aligns with its low completeness score. Similarly, mask-based methods like COB-GS [23] fail to completely filter out background pixels in complex geometries. In contrast, our BRG-Bokeh achieves the highest scores across all three dimensions and produces the visually sharpest boundaries with complete object structures, verifying that our boundary-refined strategy effectively disentangles foreground from background to ensure artifact-free bokeh rendering.

VIII. Comprehensive Ablation on Bokeh Rendering

To isolate the factors contributing to the final bokeh quality, we conduct a comprehensive ablation study on the NVOS dataset, quantitatively summarized in Table V. This evaluation is two-fold: verifying the influence of upstream segmentation quality and assessing the necessity of our tailored rendering strategies.



Fig. 4. Visual comparison of segmentation results on the “horns_left”, “flower”, and “fern” scenes from the NVOS dataset (from top to bottom).

Impact of Segmentation Quality. The top section of Table V correlates the segmentation variants #1 through #4 (defined in Table II of the main paper) with their corresponding bokeh rendering metrics. Model #1 represents the baseline without our proposed regularizations, while #4 includes background constraints ($\mathcal{L}_{bg}$), edge refinement ($\mathcal{L}_{edge}$), and depth regularization ($\mathcal{L}_{depth}$). Although the improvements in numerical segmentation metrics (mIoU/mAcc) might appear incremental in the main paper, their impact on bokeh rendering is non-negligible. As observed, the transition from #1 to #4 yields a consistent improvement in perceptual metrics, particularly reducing PIQE from 55.38 to 55.07 and increasing CLIP-IQA from 0.5624 to 0.5651. This suggests that while minor boundary noise may not drastically penalize mIoU, it manifests as visible artifacts (e.g., jagged edges or halos) in bokeh rendering. Our full segmentation model (used as “Full Method” here) further minimizes these artifacts, providing the cleanest semantic boundaries for the subsequent defocus process.

Impact of Rendering Strategies. The bottom section of Table V dissects the contribution of our three core rendering components. *Depth Reassignment* emerges as the most critical component for achieving multi-view consistency; removing it causes a catastrophic degradation across all metrics, with NIQE spiking to 6.71 and PIQE to 70.99. Without this strategy, the focal plane is determined solely by raw, often noisy depth values, preventing the target object from appearing uniformly sharp. Furthermore, *Edge Smoothing* proves essential for eliminating the unnatural “cut-out” look at object boundaries, ensuring a soft and realistic transition between the in-focus foreground and the blurred background. Finally, the omission of *Depth Jittering* negatively impacts perceptual scores, as this strategy is vital for preventing the object from appearing as a flat, artificial sticker by restoring subtle depth variations within the reassigned focal plane, thereby enhancing the geometric realism of the bokeh effect.

Qualitative Analysis of Ablation. To intuitively visualize the impact of these components, we present a qualitative

TABLE V

COMPREHENSIVE ABLATION STUDY ON NVOS DATASET. THE TOP SECTION INVESTIGATES THE IMPACT OF SEGMENTATION QUALITY (REFERENCING IDS FROM TABLE II) ON THE FINAL RENDERING. THE BOTTOM SECTION ANALYZES THE CONTRIBUTION OF EACH RENDERING COMPONENT BY REMOVING THEM INDIVIDUALLY FROM THE FULL PIPELINE (“W/O” DENOTES WITHOUT).

Configuration	NIQE↓	PIQE↓	HyperIQA↑	MUSIQ↑	CLIP-IQA↑
Impact of Segmentation Quality					
Seg. Model #1	5.62	55.38	0.5792	69.58	0.5624
Seg. Model #2	5.60	55.37	0.5783	69.42	0.5644
Seg. Model #3	5.59	55.16	0.5804	69.60	0.5648
Seg. Model #4	5.59	55.07	0.5802	69.62	0.5651
Impact of Rendering Strategies					
w/o Depth Reassign.	6.71	70.99	0.4511	57.37	0.5066
w/o Edge Smooth.	5.59	55.03	0.5801	69.61	0.5665
w/o Depth Jitter.	5.59	55.02	0.5803	69.62	0.5667
Full Method (Ours)	5.59	55.01	0.5807	69.62	0.5668

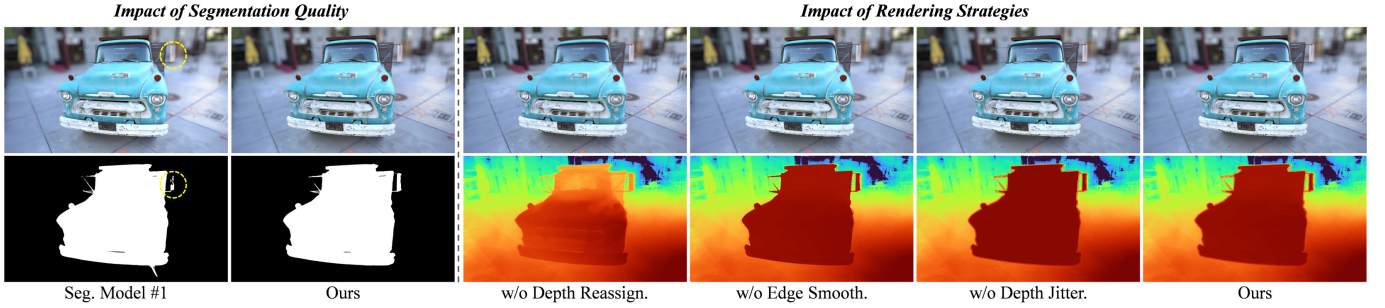


Fig. 5. Visual comparison of Ablation on Bokeh Rendering results on the “truck” scenes from the MIP-360 dataset.

comparison in Figure 5. In the *Impact of Segmentation Quality* panel (left panel), we compare the bokeh rendering results derived from the baseline Seg. Model #1 against our full method. As highlighted by the yellow circle, the baseline model fails to accurately capture the **boundary area** of the truck’s right rearview mirror, leading to **incomplete segmentation** in the predicted mask. Consequently, the rendered bokeh image exhibits focus leakage in this region, where the mirror is incorrectly blurred into the background. In contrast, our method generates a precise mask that strictly adheres to the object boundary, ensuring the entire object remains consistently in focus.

The *Impact of Rendering Strategies* panel (right panel) visualizes the necessity of our rendering components. Without *Depth Reassignment*, the rendered depth map exhibits significant variations across the object surface (visible as prominent color gradients), which leads to inconsistent focus where parts of the truck drift out of the focal plane. Removing *Edge Smoothing* results in sharp, hard transitions at the object boundaries, creating a “cut-out” appearance that lacks optical realism. Excluding *Depth Jittering* yields a completely uniform depth map for the foreground, causing the object to look artificially planar. Our full pipeline effectively balances these factors, maintaining internal depth hierarchy while ensuring boundary smoothness, thereby producing a natural and coherent bokeh effect.

Parameter Sensitivity Analysis. To determine the optimal hyperparameter settings for our rendering pipeline, we conduct a visual ablation study on the smoothing radii (r_{in}, r_{out}), the depth perturbation coefficient (t), and the lens parameter (k), as visualized in Fig. 6. Regarding the *Impact of Smooth Radii* (first row), we observe that minimal radii ($r_{in} = r_{out} = 1$) result in hard, jagged transitions at object boundaries, lacking the necessary feathering effect. Conversely, excessive smoothing ($r_{in} = r_{out} = 7$) leads to the erosion of fine geometric details at the edges and introduces visible “halo” artifacts where background colors bleed into the foreground. We empirically find that $r_{in} = r_{out} = 3$ achieves the optimal balance, effectively softening edges while preserving silhouette integrity. For the *Impact of Perturbation Coefficient* (second row), the parameter t modulates the trade-off between focus consistency and geometric relief. A small coefficient ($t = 0.15$) keeps the depth values tightly clustered around the focal plane, ensuring the entire truck remains sharply in focus. As t increases towards 0.9, the reassigned depth map progressively approaches the original raw depth distribution; while this fully restores geometric curvature, it re-introduces large depth variances, causing parts of the object distant from the focal point (e.g., the rear of the truck) to drift out of the depth-of-field and become unintentionally blurred. Thus, we select $t = 0.15$ to guarantee robust focus while avoiding a purely planar appearance. Finally, the *Impact of Lens Parameter* (third row) demonstrates the controllability of our rendering. Increasing k from 10 to 40 linearly scales the blur radius, progressively strengthening the background bokeh to

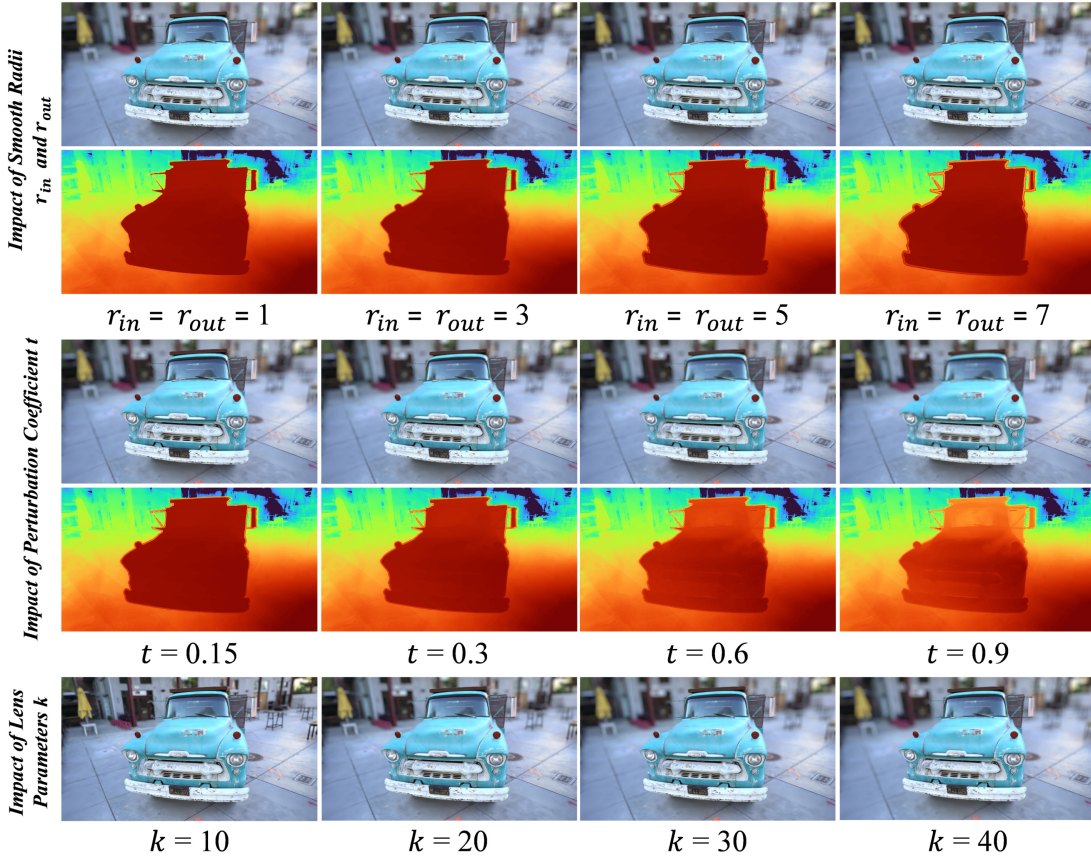


Fig. 6. Visual comparison of the parameter ablation study on smooth radii r_{in} and r_{out} , perturbation coefficient t , and lens parameter k , evaluated on the “truck” scene from the MIP-360 dataset.

isolate the subject more effectively without compromising the sharpness of the in-focus regions.

IX. Running Time

TABLE VI
RUNNING TIME.

Method \ Stage	Preparation	Training-Stage1	Training-Stage2	Rendering	Total
NeRFocus [15]	4.12min	36.44h	-	5.2min	36.56h
DoF-Gaussian [16]	2.16h	1.04h	-	23s	3.21h
DOF-GS [17]	4.12min	1.14h	-	50s	1.22h
Ours	1.83min	9.5min	11s	7.23min	0.31h

Running Time Comparison. We evaluate the computational efficiency of all methods on the “camera” scene from the DoF-NeRF [14] dataset, with all input images resized to 1/4 resolution for a fair comparison. The detailed breakdown is reported in Table VI. NeRFocus [15], as a NeRF-based method, requires extensive training time and suffers from slow volumetric rendering. DoF-Gaussian [16] incurs significant time overhead because it relies on COLMAP to extract dense depth maps for supervision, prolonging both the preparation and depth-supervised training phases. Similarly, DOF-GS [17] requires a relatively long training period to learn an “In-Focus Region Mask” to effectively model the focused regions. In contrast, our method operates directly on all-in-focus images, avoiding the complexity of learning from defocused data. Crucially, our *Training-Stage 2* (the alternating mask optimization) is extremely fast. This design offers superior flexibility: to switch the in-focus object, there is no need to retrain the entire scene; one simply needs to re-execute the brief *Training-Stage 2* and the rendering step.



Fig. 7. Visual demonstration of text-driven focus control on the “camera” scene from the DoF-NeRF dataset. The middle row shows the all-in-focus rendering. By inputting different text prompts (e.g., “Camera” vs. “The ball lamp”), our method automatically segments the corresponding object and renders multi-view consistent bokeh effects focused on the specified target.

X. Text-Driven Bokeh Rendering

Text-Driven Focus Control. We demonstrate the intuitive control capabilities of our framework on the “camera” scene from the DoF-NeRF dataset, as illustrated in Fig. 7. The middle row presents the rendered all-in-focus images across multiple viewpoints, which serve as the base for our re-focusing operations. Leveraging the open-vocabulary segmentation ability of SAM3, our method allows users to dynamically specify the in-focus subject using natural language prompts. As shown in the top row, the text prompt “Camera” triggers the system to lock the focal plane onto the camera body, rendering it sharply while applying a natural bokeh effect to the background elements (e.g., the ball lamp and string lights) consistently across all views. Conversely, by simply changing the prompt to “The ball lamp” (bottom row), the focus seamlessly shifts to the rear spherical lamp, causing the foreground camera to become out-of-focus. This experiment validates that our approach can achieve precise, multi-view consistent focus switching based solely on semantic text inputs, offering a flexible and user-friendly interaction mode.